

27 Use of matrices in multiple reactions problems

Why Are We Learning This?

In previous reactor-design problems, stoichiometry was handled using relative-rate relationships. That approach works well for simple reactions, but it becomes difficult to manage when many species participate in many reactions simultaneously. Matrices provide a systematic way to organize stoichiometric information and perform calculations involving multiple reactions. Instead of writing large numbers of relative-rate equations, we can use matrix operations to:

- Represent reaction networks compactly.
- Calculate net species rates.
- Determine heats of reaction.
- Identify independent reactions.
- Analyze complex reaction mechanisms.

These ideas will appear repeatedly in advanced reactor design, reaction mechanisms, catalytic reaction engineering, and process modeling. By the end of this lecture, you should be able to:

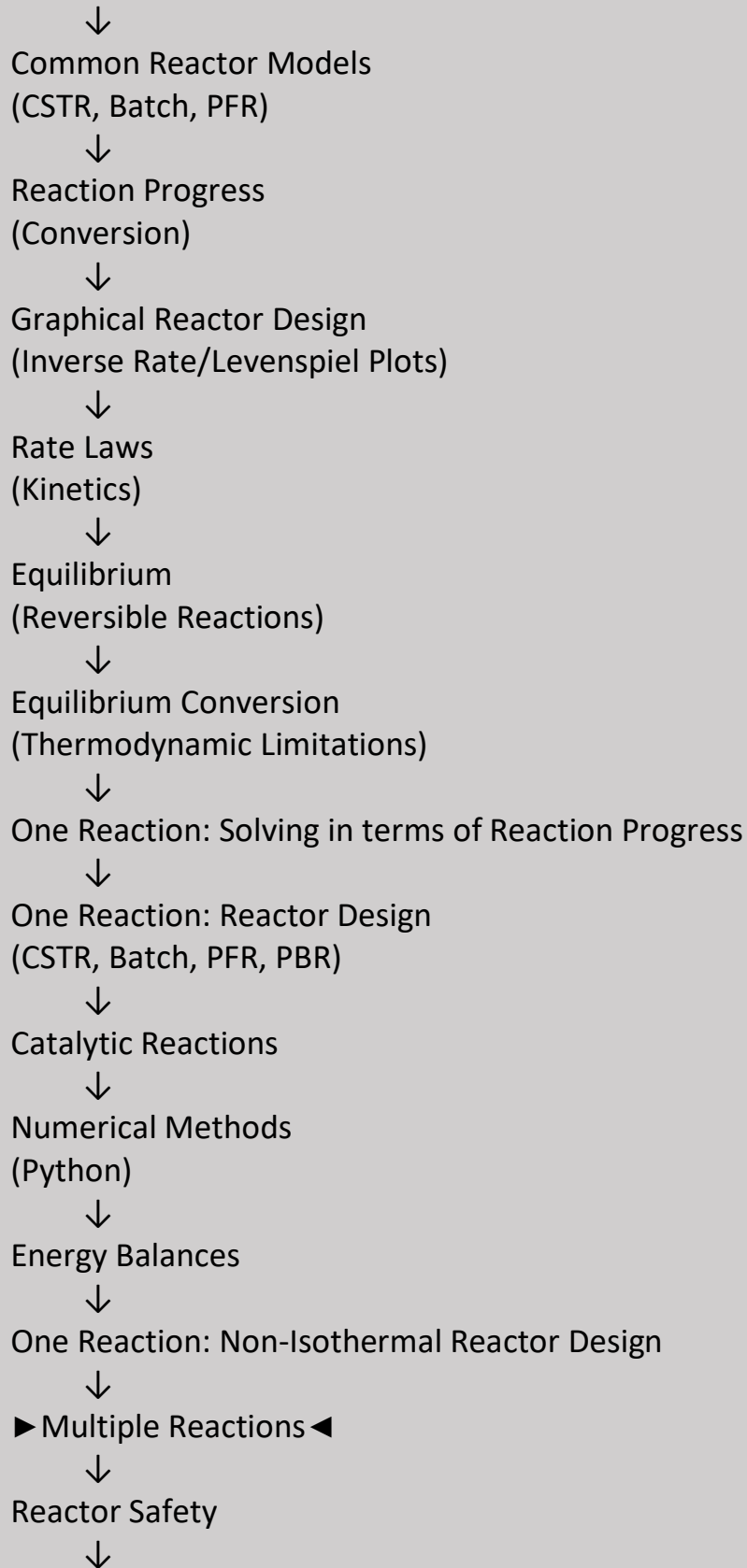
- Write chemical reactions in matrix form by constructing a **stoichiometry matrix**.
- Use the **transpose of the stoichiometry matrix** to calculate net species rates for systems involving multiple reactions.
- Determine the number of **independent reactions** in a reaction network.
- Construct a system of reactions that explains an observed product distribution.

Course Roadmap

Introduction to CRE



General Balance Equation



Challenging the Well-Mixed Assumption
(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Reaction Network



Stoichiometry Matrix



Matrix Operations



Species Rates



Reactor Design

The stoichiometry matrix turns complicated reaction networks into organized mathematical objects that are easier to analyze.

Topic 1: Writing the "stoichiometry matrix"

Essentially, this is the practice of writing a chemical reaction as

and setting it equal to

For example, consider the nitric oxide oxidation reaction, $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$.

Writing as a mathematical equation:

Setting it equal to 0:

We can recover this linear equation using matrices, i.e.,

Where A is the

In this case:

A1 =

A2 =

A3 =

is the stoichiometry matrix. Note that stoichiometric coefficients of reactants are

and stoichiometric coefficients of products are

Why Put Reactions Into Matrix Form?

A chemical reaction is really a balance between species production and consumption. For example $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ can be represented as:

Reactants = Negative Entries

Products = Positive Entries

The stoichiometry matrix stores this information in a structured form.

Sign Convention

Reactants → Negative

Products → Positive

For example: $A \rightarrow B$

would be represented as:

A : -1

B : +1

This sign convention makes matrix calculations consistent and easy to automate.

The chemical reaction written in matrix form is

For example, the stoichiometry matrix can also be used to compute ΔH^{rxn} by replacing the species matrix with a "heats of formation" matrix:

ΔC_p and really any quantity that takes a "produces minus reactants" form can be computed similarly.

The Same Matrix Can Calculate More Than Rates

The stoichiometry matrix works for any quantity that follows: Products – Reactants

For example:

Heat of Reaction

Products' Heats of Formation – Reactants' Heats of Formation

Net Production Rates

Rates of Formation – Rates of Consumption

The matrix approach provides a single framework for many reactor-engineering calculations.

The Stoichiometry Matrix Is a Database

Think of the stoichiometry matrix as a database that stores all reaction information.

Each row represents a species.

Each column represents a reaction.

The entries tell us: Produced? Consumed? How Much?

for every species in every reaction.

Why This Is Powerful

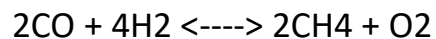
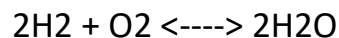
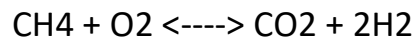
Once the stoichiometry matrix exists, many calculations become automatic.

Examples:

Net Species Rates and Reaction Enthalpies

Topic 2: Using the transpose of the stoichiometry matrix to write species' net rates

Consider the multiple reactions:



First, determine an "order" for each species in the species matrix. Let's set A1 = CH₄, A2 = O₂, A3 = CO₂, A4 = H₂, A5 = H₂O, A6 = CO, A7 = CH₄.

Since there are 3 reactions, the stoichiometry matrix has

To write the stoichiometry matrix, we tabulate the stoichiometric coefficients of these species in each reaction, i.e.,

The net rate of each species is then $v^T r$, where r are

Why Do We Care About the Transpose?

Writing the stoichiometry matrix is only the first step. The real power comes from using the transpose of the matrix to convert

Reaction Rates into Species Rates

This is exactly what reactor mole balances require.

Workflow

Reaction Network

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Stoichiometry Matrix

↓

Transpose

↓

Species Rates

↓

Mole Balances

Reaction Rates vs Species Rates

This is one of the most important concepts in reactor design.

Reaction Rates: Describe how quickly reactions occur.

r_1, r_2, r_3

Species Rates: Describe how quickly species are produced or consumed.

r_A, r_B, r_C, r_D

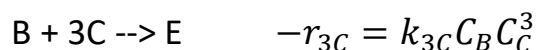
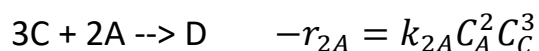
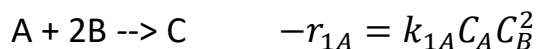
These are not generally the same. For example:

Species C: Produced in Reaction 1, Consumed in Reaction 2. Therefore:

Net Rate of C = Production – Consumption

The matrix method automatically performs these bookkeeping calculations.

An important note: Rates are defined relative to some species. For example:



The rate of reaction 1 is defined relative to species A, the rate of reaction 2 is defined relative to species A, and the rate of reaction 3 is defined relative to species C.

Common Student Mistake

Reaction rates are often defined relative to a particular species.

For example: $-r_A$ or r_C may be given rather than a generic reaction rate. Before constructing the stoichiometry matrix:

Make sure the species used to define the rate has a stoichiometric coefficient of 1. Otherwise, the matrix calculations will produce incorrect species rates.

Check Yourself

Before writing the matrix, ask: What species is this rate defined relative to?

This simple check prevents many mistakes in multiple-reaction problems.

The stoichiometry matrix must be set up so that the stoichiometric coefficient on the species that the rate is written relative to is equal to one. Using the example above:

We can then write $v^T r =$

Summary and Key Takeaways

Today we introduced matrix methods for handling stoichiometry in systems involving multiple reactions.

Key Ideas

- ✓ A stoichiometry matrix provides a compact representation of reaction networks.
- ✓ Reactants are assigned negative coefficients and products are assigned positive coefficients.
- ✓ The transpose of the stoichiometry matrix converts reaction rates into species rates.
- ✓ Matrix methods provide a systematic way to calculate net rates for large reaction networks.
- ✓ The same framework can be used to calculate heats of reaction and analyze reaction mechanisms.
- ✓ Care must be taken when rates are defined relative to a particular species.

Most Important Idea

Reaction Rates

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Stoichiometry Matrix

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Transpose Operation

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Net Species Rates

↓

Mole Balances

The stoichiometry matrix serves as the bridge between reaction kinetics and reactor design, allowing complex reaction networks to be analyzed systematically and consistently.